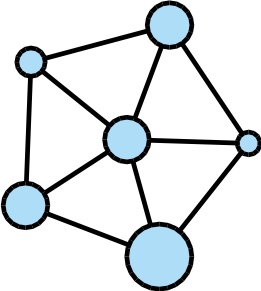
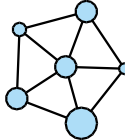
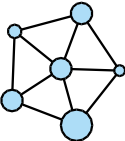
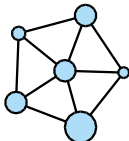


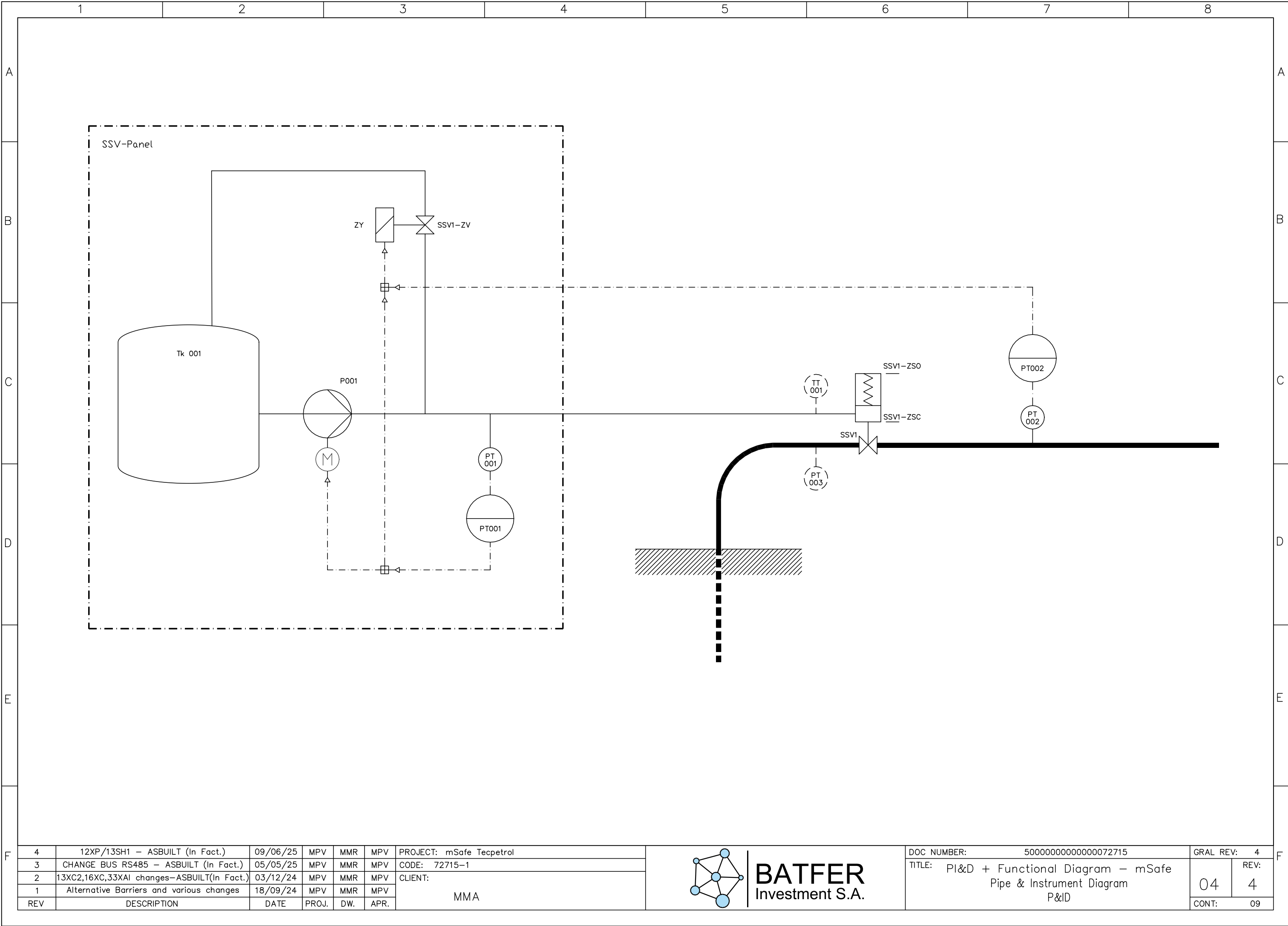
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D	Client: - MMA Project: - mSafe Tecpetrol Description Document: - 72715-1 Electrical Diagram mSafe Document Number: - 72715 ZPL Revision: - 4					APROBADO PARA PRODUCCION APROBADO: MPV FECHA: 09/06/2025  BATFER Investment S.A.			D																																				
E	Installation Place -							E																																					
F	<table><tr><td>4</td><td>12XP/13SH1 – ASBUILT (In Fact.)</td><td>09/06/25</td><td>MPV</td><td>MMR</td><td>MPV</td><td>PROJECT: mSafe Tecpetrol</td></tr><tr><td>3</td><td>CHANGE BUS RS485 – ASBUILT (In Fact.)</td><td>05/05/25</td><td>MPV</td><td>MMR</td><td>MPV</td><td>CODE: 72715–1</td></tr><tr><td>2</td><td>13XC2,16XC,33XAi changes–ASBUILT(In Fact.)</td><td>03/12/24</td><td>MPV</td><td>MMR</td><td>MPV</td><td>CLIENT:</td></tr><tr><td>1</td><td>Alternative Barriers and various changes</td><td>18/09/24</td><td>MPV</td><td>MMR</td><td>MPV</td><td>MMA</td></tr><tr><td>REV</td><td>DESCRIPTION</td><td>DATE</td><td>PROJ.</td><td>DW.</td><td>APR.</td><td></td></tr></table>					4	12XP/13SH1 – ASBUILT (In Fact.)	09/06/25	MPV	MMR	MPV	PROJECT: mSafe Tecpetrol	3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV	CODE: 72715–1	2	13XC2,16XC,33XAi changes–ASBUILT(In Fact.)	03/12/24	MPV	MMR	MPV	CLIENT:	1	Alternative Barriers and various changes	18/09/24	MPV	MMR	MPV	MMA	REV	DESCRIPTION	DATE	PROJ.	DW.	APR.		 BATFER Investment S.A.		DOC NUMBER: 50000000000000072715 TITLE: PI&D + Functional Diagram – mSafe COVER – GRAL REV: 4 REV: 4 CONT: 01		F
4	12XP/13SH1 – ASBUILT (In Fact.)	09/06/25	MPV	MMR	MPV	PROJECT: mSafe Tecpetrol																																							
3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV	CODE: 72715–1																																							
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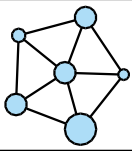
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A	Pág.	Page Title		Revision							Additional Comments	Pág.	Page Title		Revision							Additional Comments		
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02	FREE											47	FREE											
03	FREE											48	FREE											
04	P&ID	X										49	FREE											
05	FREE											50	FREE											
06	FREE											51	FREE											
07	FREE											52	FREE											
08	FREE											53	FREE											
09	Material's List	X										54	FREE											
10	FREE											55	FREE											
11	Power Generation	X	X									56	FREE											
12	Battery Chargers and Power Circuits	X				X						57	FREE											
13	Control Voltage – Generation and Distribution	X		X		X						58	FREE											
14	Control Unit – Power Supply	X										59	FREE											
15	Intrinsic Safety Barrier – Power Supply	X	X									60	FREE											
16	Common Distribution for Signals			X		X						61	Digital Outputs –Main Board	X										
17	FREE											62	FREE											
18	FREE											63	FREE											
19	FREE											64	FREE											
20	FREE											65	FREE											
21	Electronic Board (21PCB01) – Main Board	X	X			X						66	FREE											
22	FREE											67	FREE											
23	FREE											68	FREE											
24	FREE											69	FREE											
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28	FREE											73	FREE											
29	FREE											74	FREE											
30	FREE											75	FREE											
31	Inner Analog Inputs – Main Board	X				X						76	FREE											
32	External Analog Inputs – Main Board	X	X			X						77	FREE											
33	External Analog Inputs – Analog Input Module	X	X	X								78	FREE											
34	FREE											79	FREE											
35	FREE											80	FREE											
36	FREE											81	Communications	X	X		X							
37	FREE											82	FREE											
38	FREE											83	FREE											
39	FREE											84	FREE											
40	FREE											85	FREE											
41	FREE											86	FREE											
42	FREE											87	FREE											
43	External Digital Inputs – Main Board	X	X									88	FREE											
44	FREE											89	FREE											
F	4	12XP/13SH1 – ASBUILT (In Fact.)		09/06/25		MPV	MMR	MPV	PROJECT: mSafe Tecpetrol		BATFER Investment S.A.	DOC NUMBER: 5000000000000072715					GRAL REV: 4							
	3	CHANGE BUS RS485 – ASBUILT (In Fact.)		05/05/25		MPV	MMR	MPV	CODE: 72715–1			TITLE: PI&D + Functional Diagram – mSafe INDEX –					REV: 4							
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	1	Alternative Barriers and various changes		18/09/24		MPV	MMR	MPV									4							
	REV	DESCRIPTION		DATE		PROJ.	DW.	APR.									CONT: 04							

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4	12XP/13SH1 – ASBUILT (In Fact.)	09/06/25	MPV	MMR	MPV
3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV
2	13XC2,16XC,33XAi changes–ASBUILT(In Fact.)	03/12/24	MPV	MMR	MPV
1	Alternative Barriers and various changes	18/09/24	MPV	MMR	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: mSafe Tecpetrol
CODE: 72715–1
CLIENT: MMA



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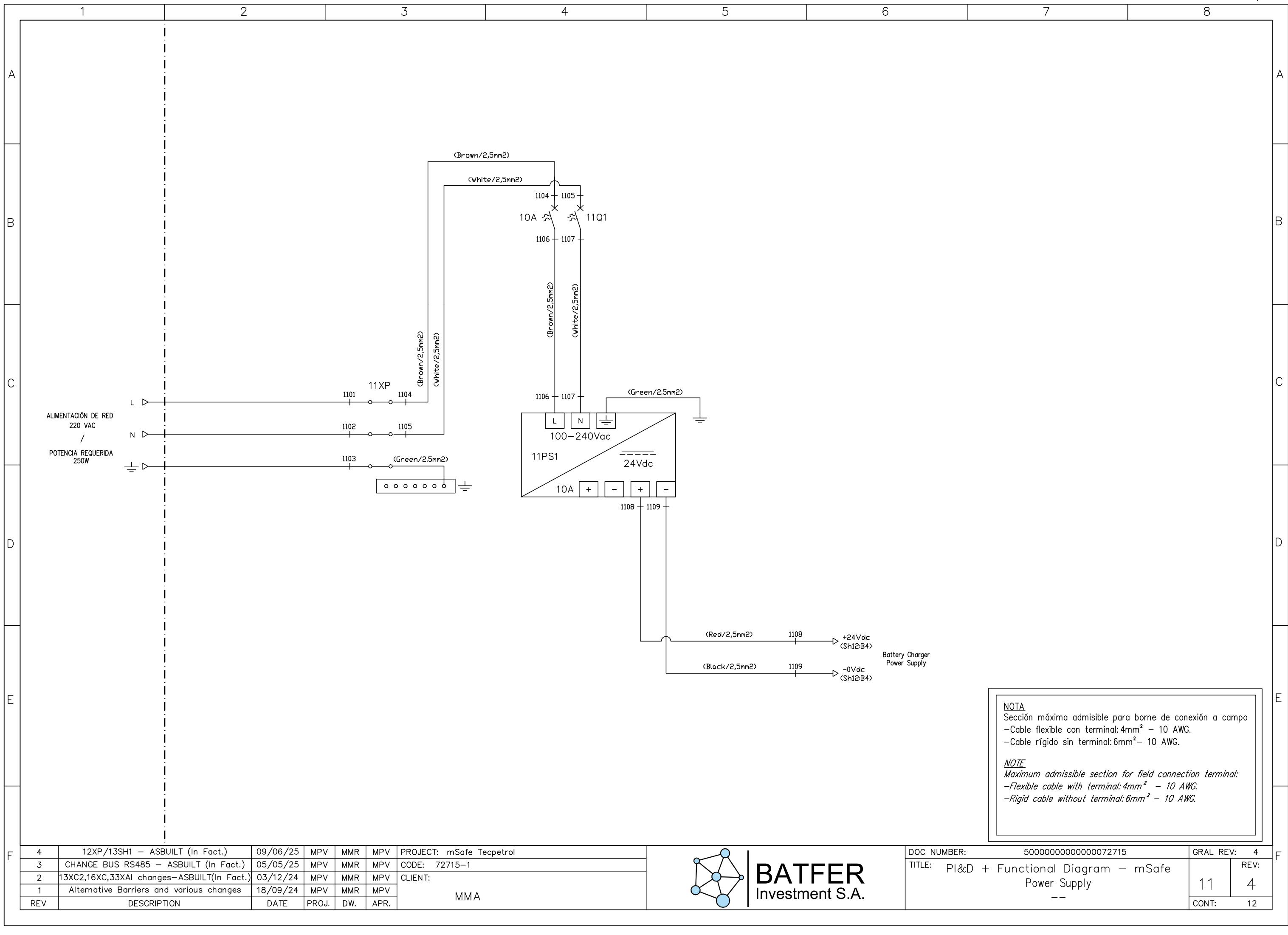
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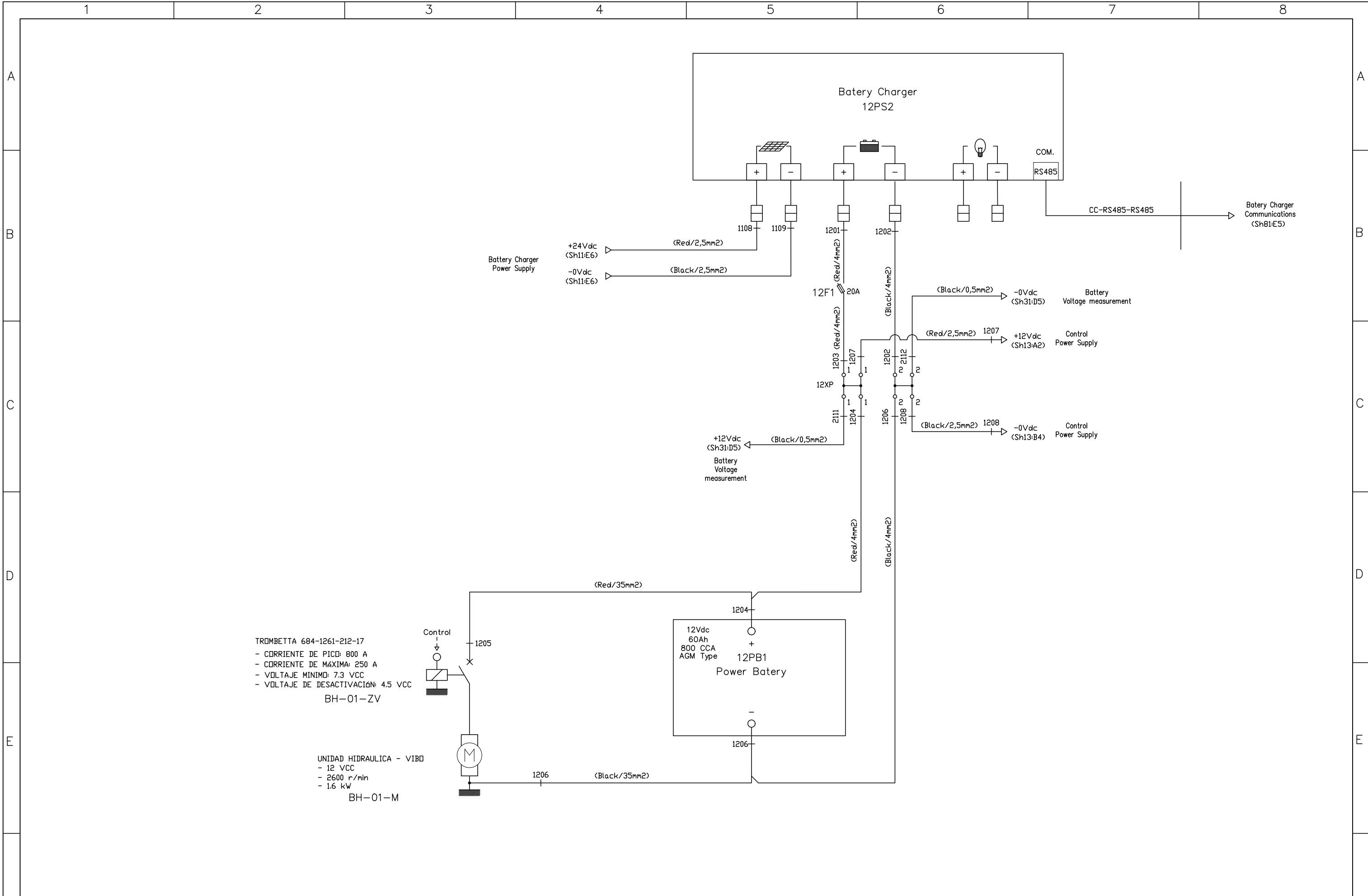
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1		2		3		4		5		6		7		8				
A	TAG		DESCRIPTION					BRAND				MODEL / CODE						
	11XP		Power Supply Terminals / Bornes de alimentación					PHOENIX CONTACT				PT 4						
	11PS1		Power Source 110–240 Vca to 12 Vdc / Fuente de alimentación 110–240 Vac a 12 Vcc					MEAN WELL				NDR–120–12						
	11Q1		AC Control Circuit Breaker / Interruptor térmico de control AC					ABB				SH 202 – C10						
	12PS2		Battery Charger 12 Vdc / Cargador de baterías 12 Vcc					EPEVER				Tracer 5210BP						
	12XP		12Vdc Power Distribution terminal blocks / Bornes de distribución alimentación 12Vcc					PHOENIX CONTACT				PT 4						
	BH–01–M		Hidraulic Pump Motor / Motor de bomba hidráulica					BREVINI				--						
	BH–01–ZV		Hidraulic Pump Motor Starter / Arrancador motor de bomba hidráulica					BREVINI				--						
B	12PB1		Power Battery / Batería de potencia					--				--						
	12F1		Power Battery Charger Fuse / Base de fusible para cargador de baterías					ABB				E 91/32						
	13XC1		12Vdc Control Voltage Distribution / Distribución de voltaje de control 12Vcc					PHOENIX CONTACT				PT 4						
	13PS2		12/24Vdc Converter / Conversor 12/24Vcc					MEAN WELL				DDR–120A–24						
	13F2		12Vdc Control Voltage Fuse Base / Base de fusible para voltage de control 12Vcc					ABB				E 91/32						
	21PCB01		Principal Control Electronic board / Placa Electrónica de Control Principal					MMA				L03–60590–01						
	31XAI		Inner Analog Inputs Terminals / Bornes de entradas analógicas internas					PHOENIX CONTACT				PTT 2,5–2MT						
	PT–001		Hidraulic Pump Pressure Transmitter / Transmisor de presión de la bomba hidráulica					GEFRAN				KS						
C	32XAI		Analog Inputs Terminals / Bornes de entradas analógicas					PHOENIX CONTACT				PTT 2,5–2MT						
	33XAI		Analog Inputs Terminals / Bornes de entradas analógicas					PHOENIX CONTACT				PTT 2,5–2MT						
	33MX01		Analog Inputs Module / Modulo de entradas analógicas					MOXA				ioLogik R1240–T						
	43XDI		External Comand 24Vdc Terminals / Bornes de comando externo 24 Vcc					PHOENIX CONTACT				PTT 2,5–2MT						
	43DIB1		Digital Input signals – Intrinsic safety barrier / Entrada Digital – Barrera Seg. Intrínseca					CHENZHU				GS8512–EX.22						
	43XDIB		Intrinsic Safety Digital Inputs Terminals / Bornes de entradas digitales Seg. Intrínseca					PHOENIX CONTACT				PTT 2,5–2 MT BU						
	43KR		Terminal Relay 24Vdc / Borne relé 24Vcc					PHOENIX CONTACT				RIF–0–RPT–24DC/21						
	61KR1..3		Terminal Relay / Borne relé					PHOENIX CONTACT				RIF–0–RPT–12DC/21						
D	XM1		Commands Terminals / Bornes de comandos					PHOENIX CONTACT				PTT 2,5–2MT						
	ZY		Solenoid Valve / Válvula solenoide					BREVINI				CRP04 NA 18W						
	81XCM		Communication Terminal / Borne de comunicación					PHOENIX CONTACT				PTT 2,5–2MT						
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4		12XP/13SH1 – ASBUILT (In Fact.)		09/06/25	MPV	MMR	MPV	PROJECT: mSafe Tecpetrol					DOC NUMBER: 5000000000000072715				GRAL REV: 4	
3		CHANGE BUS RS485 – ASBUILT (In Fact.)		05/05/25	MPV	MMR	MPV	CODE: 72715–1					TITLE: PI&D + Functional Diagram – mSafe				REV: 4	
2		13XC2,16XC,33XAI changes–ASBUILT(In Fact.)		03/12/24	MPV	MMR	MPV	CLIENT: MMA					Material's List				09	
1		Alternative Barriers and various changes		18/09/24	MPV	MMR	MPV										CONT: 11	
REV		DESCRIPTION		DATE	PROJ.	DW.	APR.											

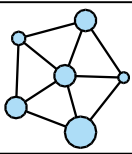
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3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV
2	13XC2,16XC,33XAI changes–ASBUILT(In Fact.)	03/12/24	MPV	MMR	MPV
1	Alternative Barriers and various changes	18/09/24	MPV	MMR	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: mSafe Tecpetrol
CODE: 72715–1
CLIENT: MMA

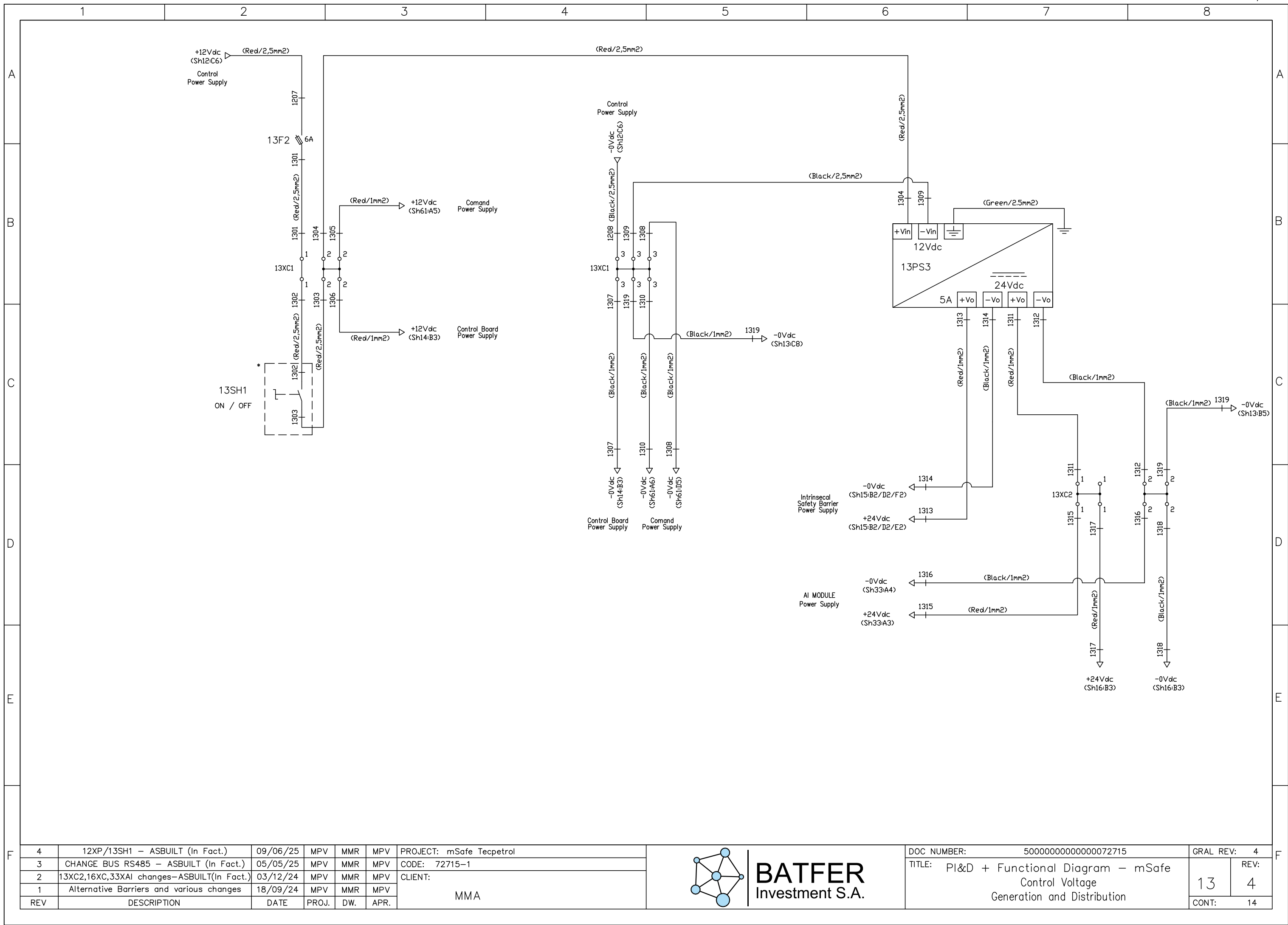


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DOC NUMBER: 5000000000000072715
TITLE: PI&D + Functional Diagram – mSafe Battery Chargers and Power Circuits
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GRAL REV: 4
REV: 4
CONT: 13

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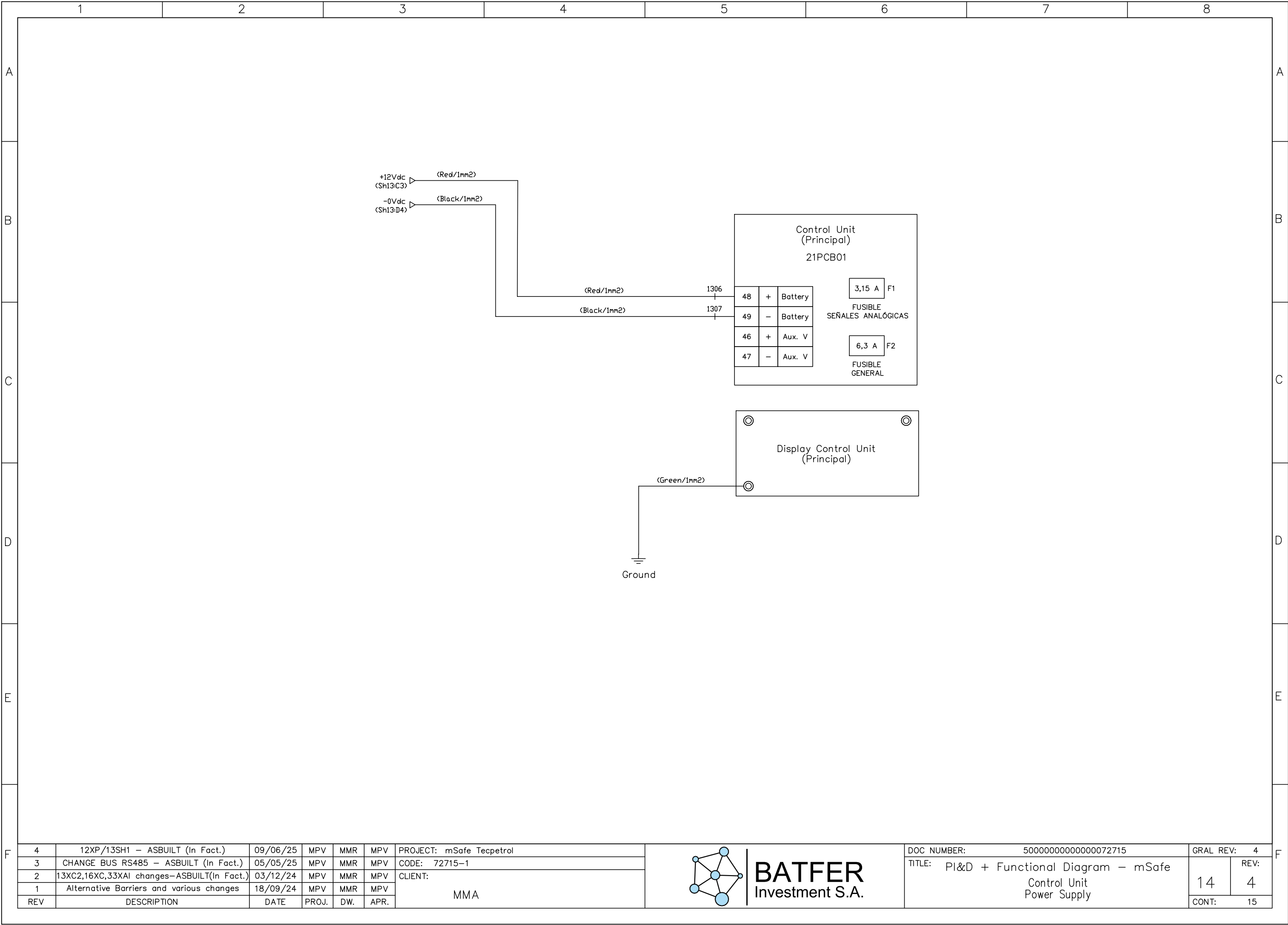
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3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV
2	13XC2,16XC,33XAI changes–ASBUILT(In Fact.)	03/12/24	MPV	MMR	MPV
1	Alternative Barriers and various changes	18/09/24	MPV	MMR	MPV
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.

PROJECT: mSafe Tecpetrol
CODE: 72715–1
CLIENT: MMA

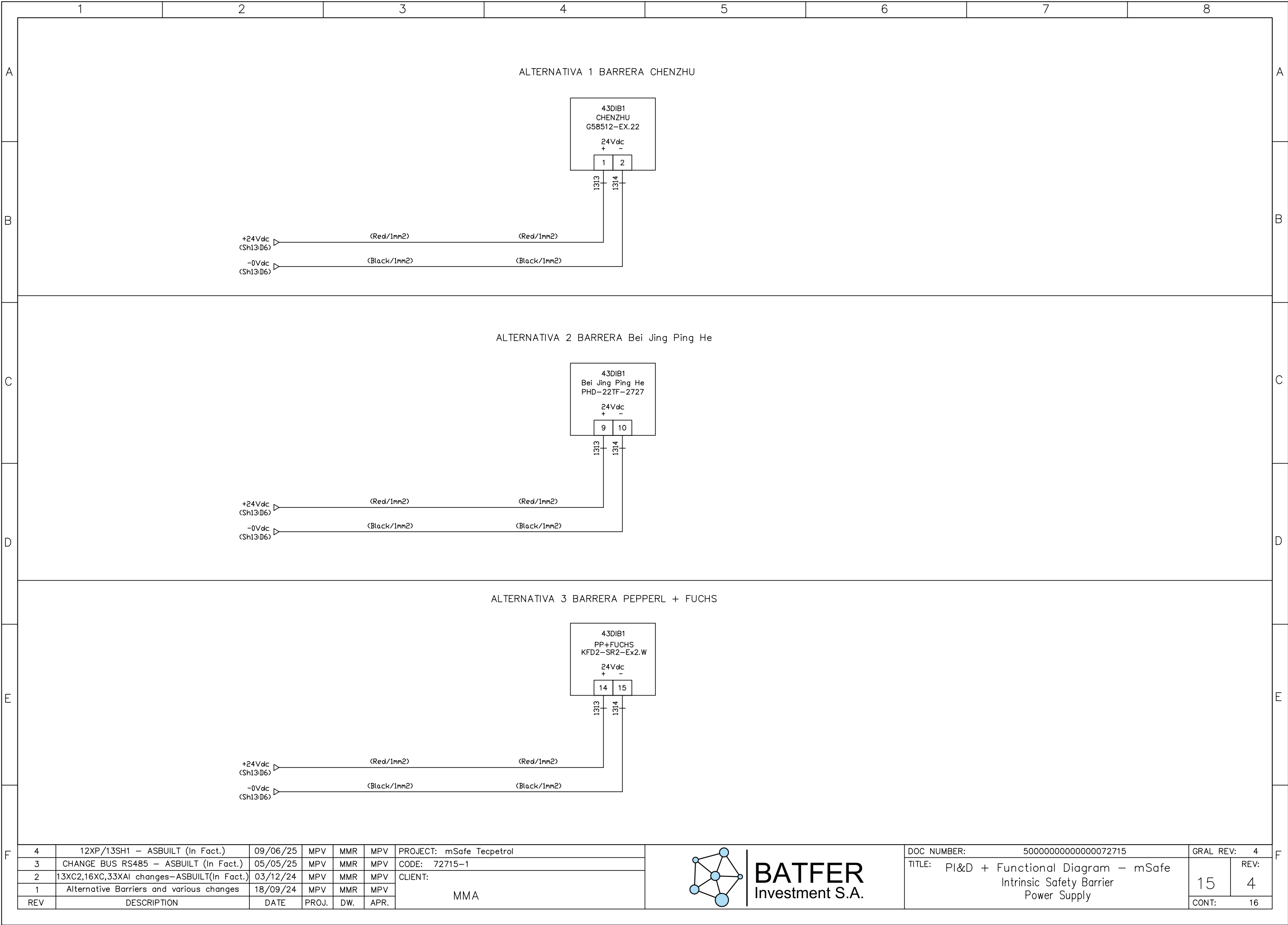


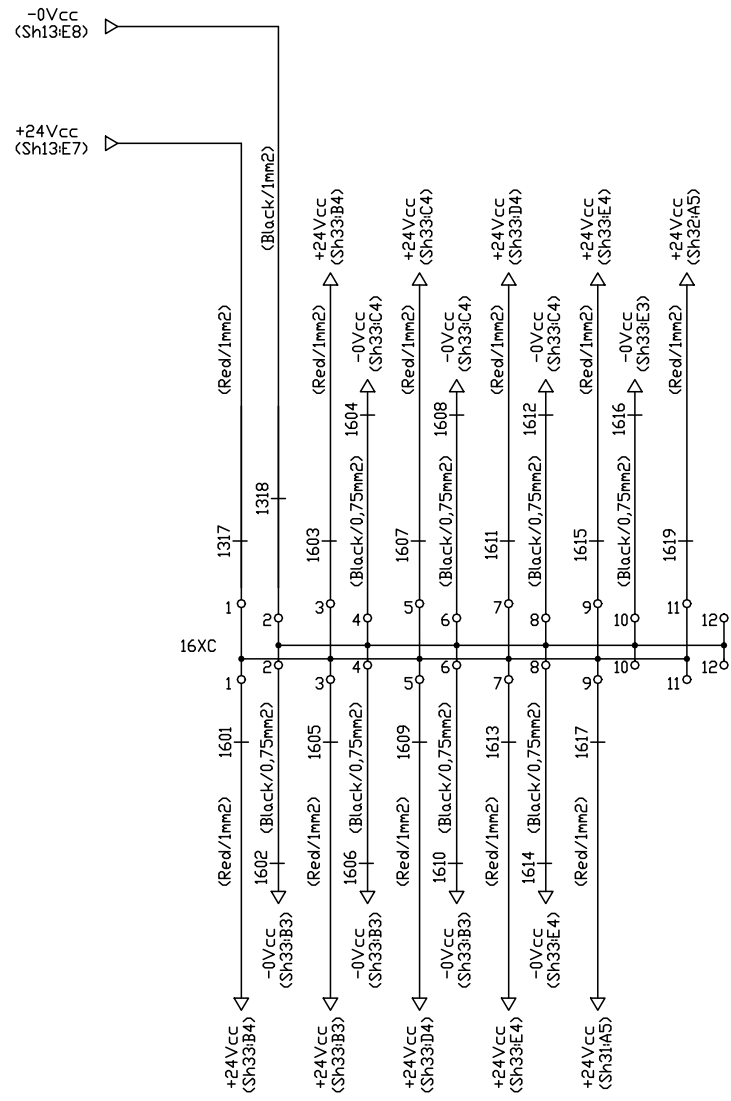
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TITLE: PI&D + Functional Diagram – mSafe Control Voltage Generation and Distribution

GRAL REV: 4	REV: 4
CONT: 14	

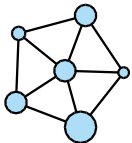


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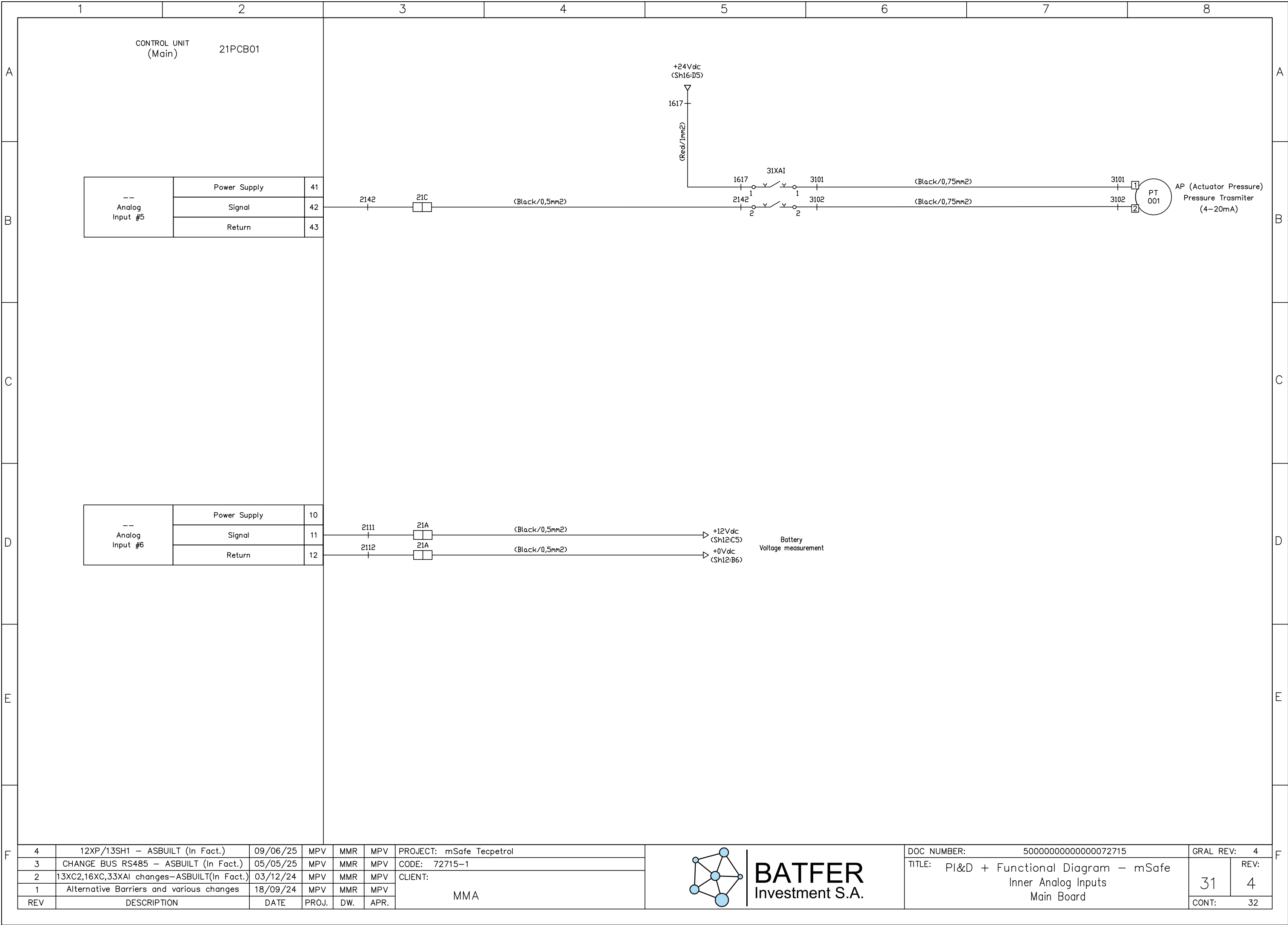




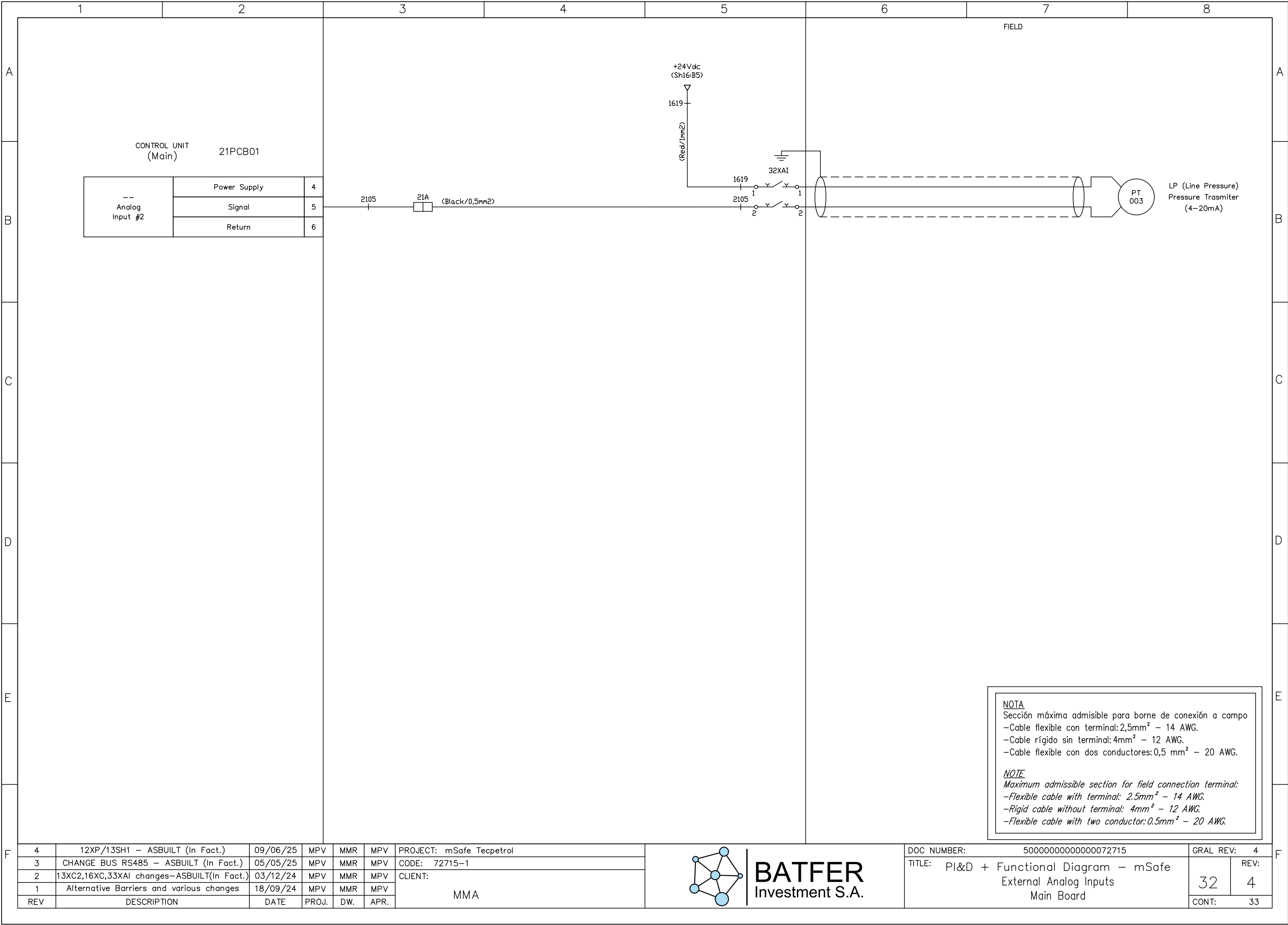
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3	CHANGE BUS RS485 – ASBUILT (In Fact.)	05/05/25	MPV	MMR	MPV	CODE: 72715–1		TITLE: PI&D + Functional Diagram – mSafe Common Distribution for Signals	16	REV: 4	
2	13XC2,16XC,33XAI changes–ASBUILT(In Fact.)	03/12/24	MPV	MMR	MPV	CLIENT: MMA					
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REV	DESCRIPTION	DATE	PROJ.	DW.	APR.			CONT: 21			



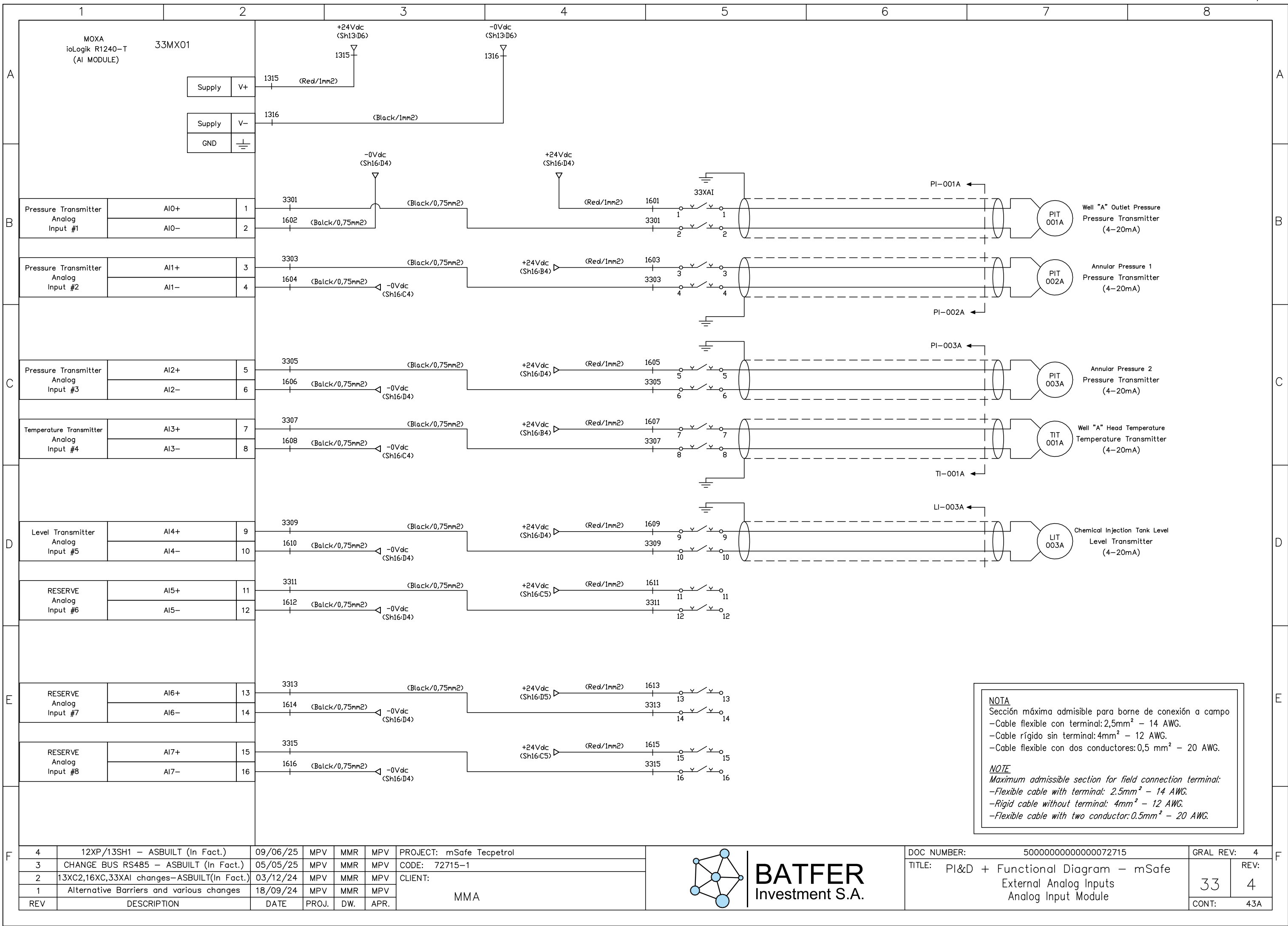
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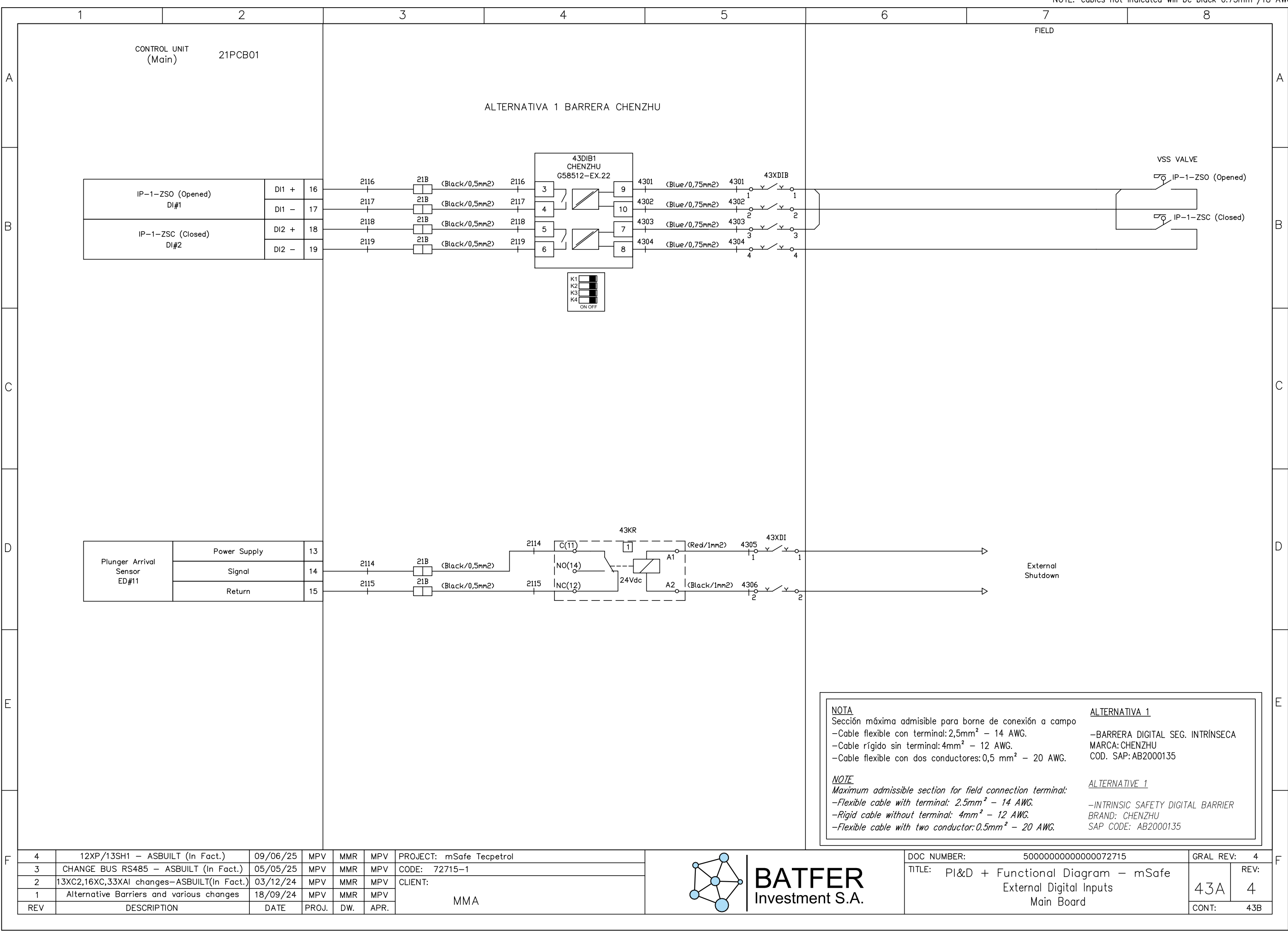
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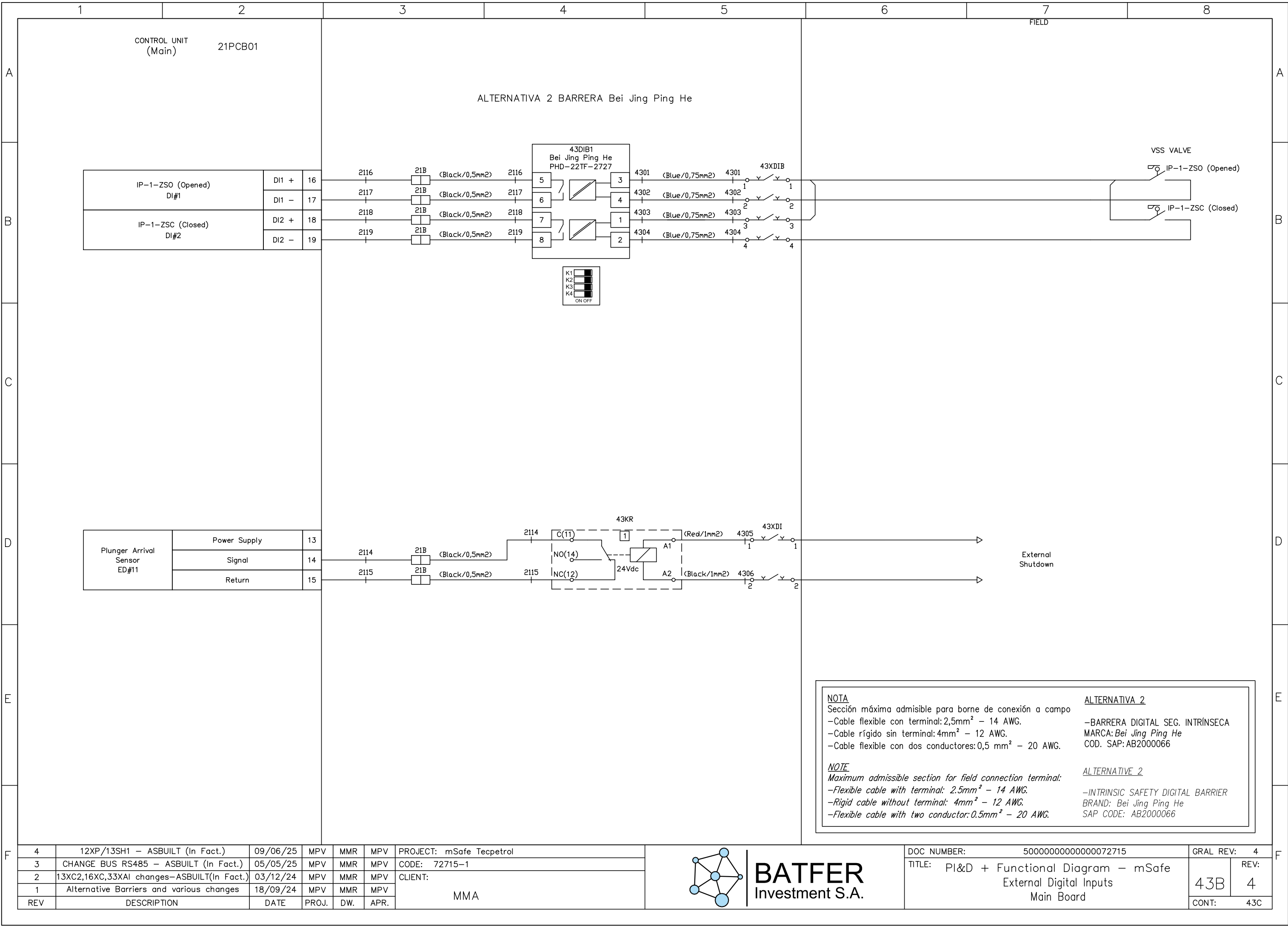


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NOTE: cables not indicated will be black 0.75mm²/18 AWG



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